

capacity will shoot up, and we may see the emergence of a graphics card that's beyond our current imagination of a mobile phone.

Around the same time, the touch pad will become *de rigueur* for most phones. A graphics artist will have powerful image applications on his phone, the compulsive document creator will be able to create presentations on the fly (you can do that even now, but with certain restrictions), and the field executive will have all the software needed for such things as e-mail and contact information. Good-bye laptop, hello smartphone.



By this time, voice would be the default way to dial, not just numbers, but also Web sites and e-mail addresses. The phone cradle will evolve into a docking station capable of communicating via Bluetooth, with all kinds of devices: keyboards, mice, monitors, and so on. The device will have expansion slots or adapters for memory, other I/O devices such as a touchpad, tablet, etc, an external camera, and a game console. A pull-out module will have a keyboard and trackball, and there should be a slot for a laser projection keyboard.

A large hard disk capacity should allow users to record TV programmes as well as video-conferences conducted using the phone. With two displays, it will be possible to do more than one thing at once—for example, talking while surfing, or talking while jotting down notes.

Quad-band phones

Phones by then would have become quad-band, a development that would not raise any eye-brows whatsoever. A new chip allows phones to choose what protocol—WiFi or 3G, for instance—is best suited for communication in a certain area; such chips will become routine, and phones will automatically select the network that's best to talk to another device. Synchronising with PCs will be seamless, and input will be Bluetooth-enabled and seamless too: phones will feature a wireless Bluetooth-connected smart pen for handwriting-recognition, and a cordless headset.

The big question when it comes to entertainment will be: will personal entertainment



The Indian Connection

It will not be long before India hops on to the smartphone bandwagon. Officially, that is. Even though it's discouraged by authorities, the grey market still brings in most of the latest products from across the world. For instance, the Sony Ericsson P900 and the PalmOne Treo 600—both among the best in the currently available smartphone category—were already available in the grey market before they were launched officially.

Officially, though, look out for the Panasonic X700, which will be launched simultaneously in India and Europe. LG is not far either: It recently launched the G7100, the "first ever camera phone with a 270-degree spin screen and a slide-mounted VGA camera", according to P Valecha, Product Group Head, mobile phones, LG Electronics India.

Harit Nagpal, chief marketing officer, Hutch, says it is a given fact that smartphones will make it big in India. "The smartphone category is a niche, albeit a growing one, in India. The mobile will become the single access device delivering a host of wireless services—whether it is for voice telephony, a stock trading desk, a game console, whatever you want. We are seeing a movement towards a single access device."



Sony Ericsson P900

devices become obsolete if phones become a complete audio-video solution, suitable for audiophiles and videophiles? A device that can do this will double up as an MP3 player and FM radio, iPod-style.

The second display will be a high-resolution one, for TV, games, and so on. The camera, meanwhile, will be capable of up to 6 megapixels (and more, later) by end-2005. There will be a slew of models that feature videocams, and these will have an extra flip for the videocam preview. High-quality stereo speakers will complete the audio-video subsystem.

Scanners will also become part of the phone, to help people instantly scan telephone numbers, names, URLs and e-mail addresses, in addition to barcodes. Future smartphones may also incorporate SIM chips that can be used to scan and purchase items from supermarkets, with the correct accounts automatically being debited or credited.

By then, the phone would have become part of our clothing, and that would make convergence take on a whole new meaning! ■

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You are going to have the equivalent of a persistent T1 line in your pocket. That's it. It's going to happen

Jeff Hawkins
Inventor of Palm Pilot



What and When... Digit peeks into the future

Late 2004

- ❑ Camera resolution and focus will improve, and the 3-megapixel camera would make an appearance. Samsung already has a prototype out.
- ❑ Phones will double up as MP3 players and FM radios, with hi-power outlets for headphones.

Mid 2005

- ❑ Videocams will become part of standard equipment.
- ❑ Phones will be equipped with scanners that can read phone numbers,

business cards, e-mail addresses and URLs, and dial them.

- ❑ If RFID takes off, its modules might be incorporated into smartphones, and you will be able to read barcodes with your phone—and make purchases from supermarkets with the accounts being debited and credited seamlessly.

Late 2005

- ❑ Speech modules will be more comprehensive, allowing for speech-based typing, and

indeed, everything that's speech-based.

- ❑ Applications will become more advanced; the move will be towards 'more of a computer', with hard disks storing all data.
- ❑ Pull-out modules with keyboards and trackballs will become more common.
- ❑ Chips that seamlessly move between wireless and cellular protocols will become ubiquitous.

Mid 2006

- ❑ Fingerprinting technology to identify

the user of the phone and preventing theft will be more advanced; currently, these systems can easily be foiled with the right knowledge.

- ❑ GPS connectivity will become a more universally available option, with maps being displayed on the phone that aid in driving and even walking to your destination.

Late 2006

- ❑ 3D displays will become common.
- ❑ Most phones will come with a cordless earpiece-

mouthpiece combo.

- ❑ Remote controlling of household appliances will be a feature that several models offer.
- ❑ The phone will be closer to replacing laptops—the device will dock to peripherals wirelessly, a la laptops, through a docking station. Laptop use will have reduced dramatically.

2007

- ❑ Advanced image-editing and other such software will appear. Virtual, or projection keyboards will be standard.